

India is at the forefront of design owing to the cutting-edge chip design activities taking place in multinational design companies that have large-scale engineering operations here.”

—Sanjay Mittal, managing director, Yogasa Systems



have matured and appear headed for single-digit growth in coming years, compared to the double-digit growth of yesteryears. However, the scene in the Indian semiconductor industry is quite different. Rising middle-class income levels have created a huge market for semiconductor-intensive products and we may see this trend to continue for some more years.

According to Kumar, globally, the number of new design start is declining, whereas the value of each new design is increasing mainly due to the continuing trend of system-on-chip integration.

Amid this recession, there is a great opportunity coming up for low-cost electronic products. The Indian electronics industry is bracing up to take on the opportunity with a few advantages it already has. India, however, remains a major importer of electronic materials, components and finished equipment amounting to over \$12 billion in last five years. The electronics hardware output of India is worth \$11.6 billion at present. The Indian electronics industry thus has the resilience to withstand the recession.

Partnering for success

Among the more enduring examples is the partnership between Nortel and Wipro, which goes back 20 years. In 1991 Nortel set up an offshore development centre (ODC) at Wipro to develop and support telecom products for service providers and enterprise customers worldwide. Today, more than a thousand engineers at Wipro

work on Nortel projects.

Nortel has since partnered with many more Indian design companies (including Infosys, Sasken and TCS), fueled by its initial success. Wipro has also benefited from the success of the Nortel ODC and now has similar centres for such clients as Ericsson and Nokia.

More recently, Japan's Renesas Technology set up its first ODC at KPIT Cummins Infosystems to strengthen its design and development capabilities. KPIT Cummins previously executed design projects for Renesas in several areas, including SoC products for use in digital consumer electronics, analogue circuit designs and embedded software.

Brand India

India is well-recognised in the international chip design services market for its technical skill, domain knowledge and access to a large talent pool.

Buoyed by these strengths, the sector is growing at almost twice the rate of application software. There are currently around 150 chip design and 100

embedded system players in the ecosystem and more and more companies are eyeing this potential market.

“Indian companies are migrating from being centres of resource augmentation to product innovation and design centres. The last few years have seen a huge growth in the number of small and innovative companies doing complete design of innovative embedded products. A similar trend exists in the captive design centres of multinationals and Indian service companies. These are becoming accountable for more and more of complete product design/delivery/ownership,” says Praveen K. Ganapathy, director-DSP business development, Texas Instruments India.

Access to local customers driven by domestic consumption will only enhance product knowledge and further catalyse the maturation of India companies to be able to do complete product development, drive product strategies and roadmaps, and do high-end consulting.

EDA solutions ‘made in India’: Fact or fiction?

Advances in electronic design automation (EDA) technology have barely kept pace with the increasing complexity in IC designs and growth in the semiconductor industry. Will we ever see a ‘made in India’ tag for EDA products and services?

Recent focus in the EDA industry has been on topics that affect manufacturing and yield, such as DFM and SSTA. Such topics warrant a deeper understanding of the semiconductor

“Increasingly, companies are realising that cost arbitrage will take them only to a certain level, and thus they are seeking avenues for ‘value’ creation with innovative designs.”



—Rajiv Batra, managing director, Rabyte